Music as a Potential Distractor: Does Language Comprehension of Music Affect Reading Comprehension?

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Abstract

Studying with background music is a common habit among college students, however, it remains unclear whether understanding the language of the music interferes with reading comprehension. This study examined whether linguistically comprehensible versus incomprehensible music had differing effects on reading performance. Participants read three passages under three distinct conditions: music in English, music in Swedish, and no music. Eye movements were tracked and monitored utilizing eye-tracking technology. Reading comprehension was measured utilizing brief quizzes at the end of each trial. No significant differences were observed for average response time, fixation count, dwell time, or blink rate. However, average eye amplitude revealed a difference approaching significance. Follow up analysis revealed a significant difference in eye amplitude between English and Swedish music conditions, suggesting that comprehensible lyrics may slightly alter attention during reading. Overall, the current findings suggest that while readers may seemingly maintain comprehension accuracy, listening to music in a comprehensible language may increase cognitive effort while reading.

Keywords: Reading comprehension, attention, comprehensible language, eye-tracking, eye amplitude

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Introduction

It has become quite common for college students to listen to music while studying. In fact, walking into any library, coffee shop, or typical study spot–it is almost fully guaranteed that some individual will be spotted utilizing headphones or earbuds. Many people listen to music while working or studying under the impression that music helps increase concentration and even block out potential distractions in one's surroundings. Despite the commonality with which studying is often paired with listening to music, the effectiveness of such a habit, especially during reading-based tasks, may not produce as positive of an effect as most people tend to think. Since reading comprehension largely depends on attention and language processing, background music may inhibit a person’s ability to effectively comprehend reading content. This has large implications on academic performance, and effective studying habits and cognitive processing.

One factor that was of particular interest while inquiring about the effects of music, was whether the listener’s ability to understand the language of the background music may affect reading comprehension. For instance, music with comprehensible lyrics may be competing for linguistic processing resources that are utilized for comprehending the reading material. Under this logic, listening to music in an unfamiliar language while reading may function more similarly to reading without background music.

Pinpointing how and to what extent language comprehension of background music affects reading could provide insight to identifying more useful studying habits–which would especially be useful for students who frequently engage in heavy reading-focused academic work. Not only would this be useful for college students, but for students of any age, and may particularly help set up adolescence for success during their formative years of schooling by instilling more effective studying habits early on that aid with memory retention and help with consolidating information.

Previous literature has examined how listening to music affects reading comprehension by analyzing readers’ eye movements during reading tasks. For example, Zhang, Miller, Cleveland, and Cortina (2018) used eye-tracking methods to investigate how college students read academic passages while listening to music compared to while reading in silence. Their findings indicated that while basic word recognition remained largely intact, reading with music led to increased rereading to compensate for worse comprehension of said reading, and longer total viewing time. This indicates greater processing difficulty. These results suggest that music may serve as a distractor while reading through introducing competition for attention, even though outcomes are not drastically impaired. However, there is also research examining the effects of background music on cognitive performance that produced mixed findings. Furnham and Stephenson (2007) investigated whether background music interferes with tasks such as reading comprehension, verbal reasoning, and mental arithmetic in young school children. Across multiple music conditions and cognitive tasks, no consistent evidence was found indicating that background music significantly impaired performance compared to silence. These findings suggest that background music alone may not be enough to disrupt cognitive performance. This highlights the gravity of identifying additional factors at play, such as the particular type of task demanded or characteristics of the music (genre, tempo, volume) which may contribute to distraction. One factor that potentially influences the distracting effects of background music, is whether the listener understands the language said music is in. Abril and Flowers (2007) for one, examined how song language affects attention and distraction in middle school students with different linguistic backgrounds. They found that music with lyrics produced more self-reported distractions than instrumental music, and that monolingual students listening to songs in an unfamiliar language (Spanish) reported distraction patterns more alike to instrumental music than to music in their native language (English). These findings suggest that linguistic comprehension plays a critical role in determining how music captures attention, with comprehensible lyrics posing a greater risk of distraction than unfamiliar music. Overall, these three studies indicate that while background music may not uniformly impair cognitive performance, music with comprehensible lyrics may act as a distractor producing inhibiting effects on a given task necessitating linguistic processing (such as reading).

The study at hand examined whether understanding the language of background music affects reading comprehension. It was hypothesized that under the condition that participants listen to a song in a language they understand while reading, their comprehension of said passage will decrease. Participants in the study completed reading tasks under three distinct conditions to test this hypothesis: listening to music in English (a familiar, known language for all participants involved), listening to music in an unfamiliar language (in this case Swedish was chosen), and as a control–reading in silence with no music playing. It was expected that reading comprehension and attentional focus would be diminished in the English music condition compared to the Swedish music or no music conditions due to increased linguistic interference. Furthermore, the Swedish music condition was expected to produce less interference compared to the English music condition since participants were unlikely to understand the lyrics, and hence produce a similar outcome to the no music condition. To test these hypotheses, participants' eye movements were monitored using an eye-tracking device while reading different text passages for each condition. Reading comprehension was further assessed using short quizzes administered at the end of each of the three reading tasks.

Method

Participants

The sample for this study consisted of 47 participants, ages 18 years and older from the Connecticut College student population. Originally, there were 50 participants, but 3 data sets were left out of the statistical analysis due to lost data, but were included for demographics. All participants were recruited either through word of mouth or through the SONA study posting online. Participants were compensated with 30 minutes (0.5 hours) of research hours for participating in the study.

Stimuli

For this experiment, three music conditions were chosen to be randomly paired with three reading passages of similar length and complexity. The three conditions include music in a different language, music in English, and silence. For the condition with foreign music, the song Regnet by Melby in Swedish was chosen. Swedish was chosen with the intention to minimize the likelihood of participants recognizing the language. For the condition with English music, the song Heartbreaker by CHERUB was selected. Both songs were found on Youtube and were intentionally chosen to be at the same bpm and pitch. Each of the three reading passages chosen are at a twelfth grade reading level and presented in a random order. One passage discussed vaccines and autism, another discussed concussion rates and effects in sports, and finally the last discussed the morality of euthanasia. The reading passages were taken from the website: https://englishforeveryone.org/Topics/Reading-Comprehension.html.

Design

The study involved a within-subjects design where each participant experienced all three experimental conditions. As stated above, the different auditory conditions were music in English, music in Swedish, and a no music or silent condition. Each participant experienced the auditory conditions in a random order, paired with one of three randomly selected passages. Passages and audios were randomly selected using MatLab–the software utilized to code said experimental conditions. The code also controlled the Eyelink 1000–the eyetracker–so the eyetracking data was correctly paired with passages and songs seen during the different trials. The Eyelink 1000 uses a sampling rate of 1000 Hz, and the experiment was presented with custom code created by Dr. Jeff Moher using Matlab and Psychtoolbox. The passages were used as a way to track participants' reading comprehension during the different auditory conditions. Eyetracking data and the results from the reading quizzes were used to assess where participants looked while reading and if they understood that content.

Procedure

Before the experiment officially began, participants were given an informed consent sheet and an opportunity to ask any questions or concerns that they may have had. Once signed, participants were set up in the eyetracker and instructed on how to use it: rest chin on the pad, forehead against the brace, and try not to move while reading the passages. The participant had to complete calibration and validation tests on the eye tracker to ensure accuracy and stability of the tracking on the pupil. Upon completion of these tasks, the experiment officially began. There were three trials conducted, each one with a different passage and a different auditory condition, which consisted of music in English, music in Swedish, or complete silence. Each of these stimuli were randomly selected for each trial to minimize any unintentional effect it may have on the participants' data. These passages were read without the experimenter present. When a participant completed the passage, they were told a number to give to the experimenter, who would then administer a comprehension quiz that corresponded to said number. Participants were told about the quizzes in advance, and took them on a computer separate from the eye tracker. This was repeated two more times with the other conditions. After the completion of all three passages and quizzes, the participants filled out a demographics questionnaire that asked them about their study habits (particularly around background sound), the languages they knew, and whether or not they recognized any music used in the study. This questionnaire was the final part of the experiment, and the participants were given debriefing forms afterwards.

Results

Looking at the Google Form participants took at the end of the experiment, we were able to see if there were any confounding variables that would affect the results of the experiment. We found that 15 of the participants spoke languages at home that were not English, and of those 15, 9 of them did not speak English as their first language. None of the participants spoke Swedish and none of them recognized either of the songs. Preference of studying conditions was also polled and it was found that there were many different ways in which people study–with no music or sound being the most common, reported by 14 of the 50 participants. The pass rates of the Google Forms taken at the end of each passage were: 98% for Passage 1, 100% for Passage 2, and 97% for Passage 3.

ANOVA tests were conducted to investigate the differences in average amplitude, average response time, average fixations, average dwell time, and blink rate.

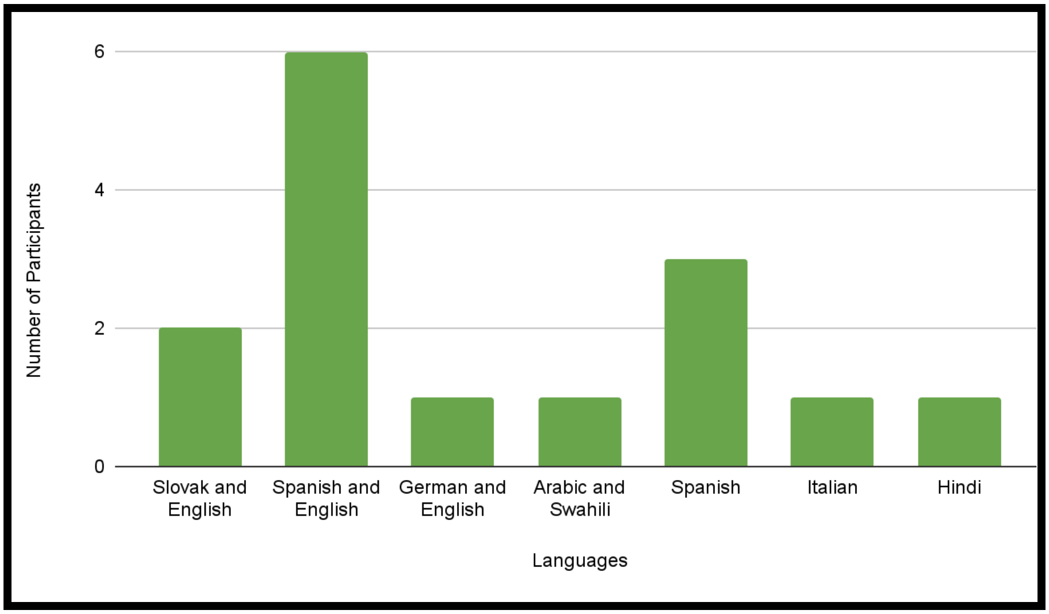

Figure 1. Languages spoken at home for participants who did not exclusively speak English in their households.

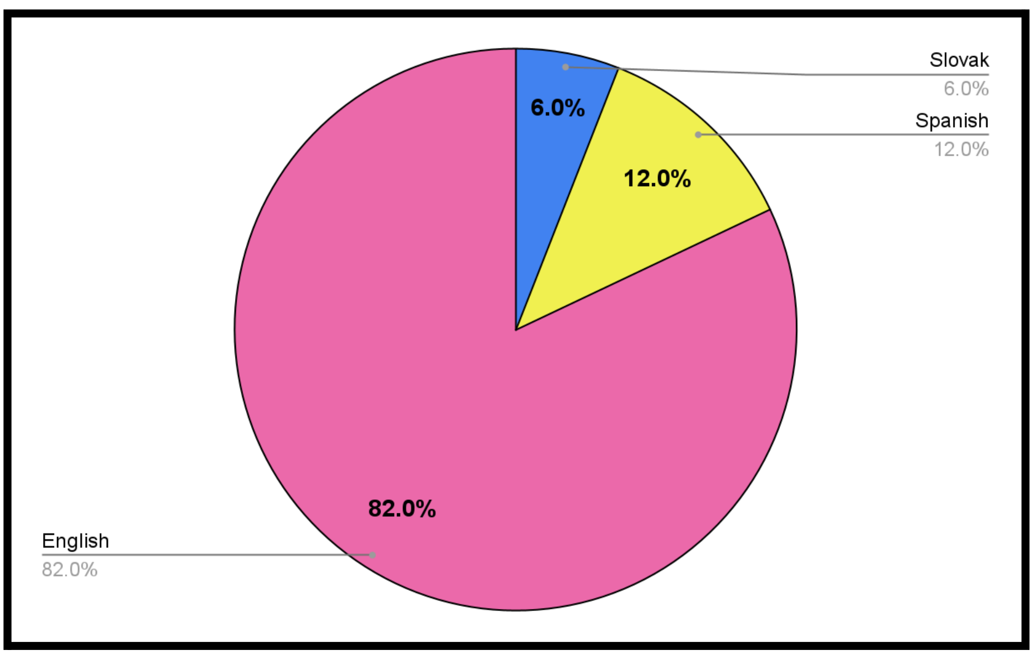

Figure 2. Demonstrates the percent of first or primary language of participants.

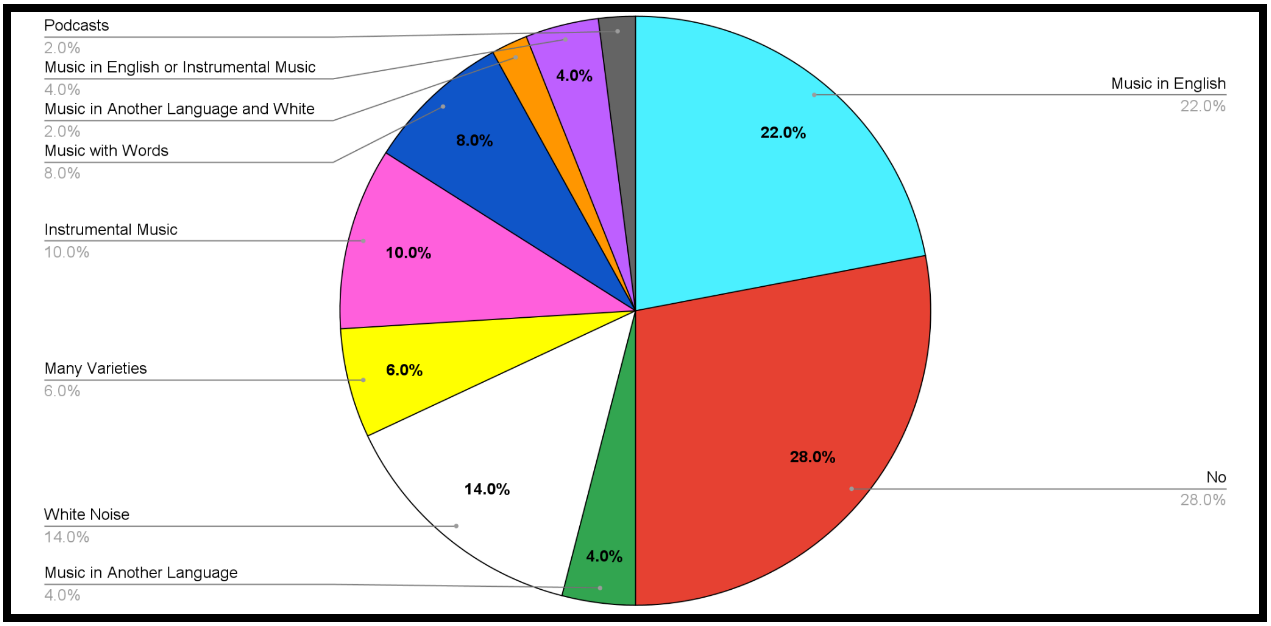

Figure 3. Shows the post-quiz survey data on what people listen to while they study.

By conducting a one way ANOVA test, we found a difference that was approaching significance of the average eye amplitudes of participants across the English, Swedish, and no music modalities, $F(2, 92) = 3.016$, $p = 0.054$. A paired samples t-test was conducted to investigate whether there would be a significant difference between each condition. We found that there was a significant difference between the average eye amplitude of the English and Swedish modalities ($M = 0.436$, $SD = 1.40$), $t(47) = 2.16$, $p = 0.036$. We found that there was no significant difference between the average eye amplitude of the English and no music modalities ($M = 0.323$, $SD = 1.48$), $t(47) = 1.52$, $p = 0.137$. We found that there was no significant difference between the Swedish and no music modalities ($M = -0.113$, $SD = 0.732$), $t(47) = -1.07$. This suggests that listening to music while reading in general could potentially have an effect on eye amplitude, but that the language of the music does in fact have a significant effect on eye amplitude when looking specifically at English versus Swedish music.

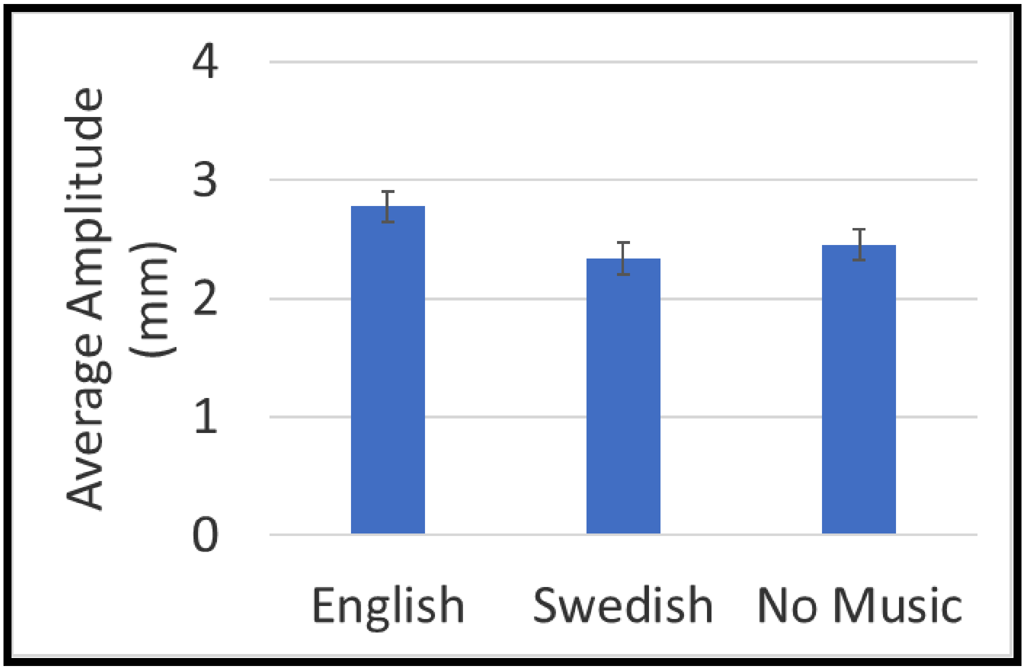

Figure 4. The average amplitude for all three conditions. Error bars reflect standard error of the mean.

By conducting a one way ANOVA test, we found no significant difference in the average response times of participants across the English, Swedish, and no music modalities, $F(2, 92) = 1.184$, $p = 0.311$. This suggests that listening to music, whether in English or Swedish, while reading has no significant effect on the average response time of participants.

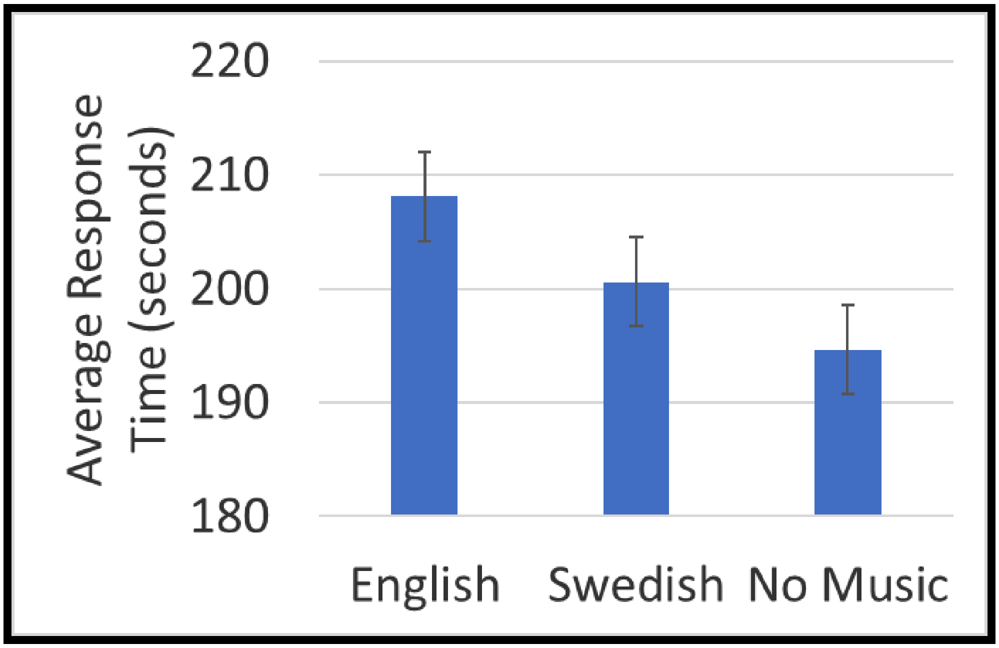

Figure 5. The average response time for all three conditions. Error bars reflect standard error of the mean.

By conducting a one way ANOVA test, we found no significant difference in the average eye fixations of participants across the English, Swedish, and no music modalities, $F(2, 92) = 0.601$, $p = 0.550$. This suggests that listening to music, whether in English or Swedish, while reading has no significant effect on the average eye fixations of participants.

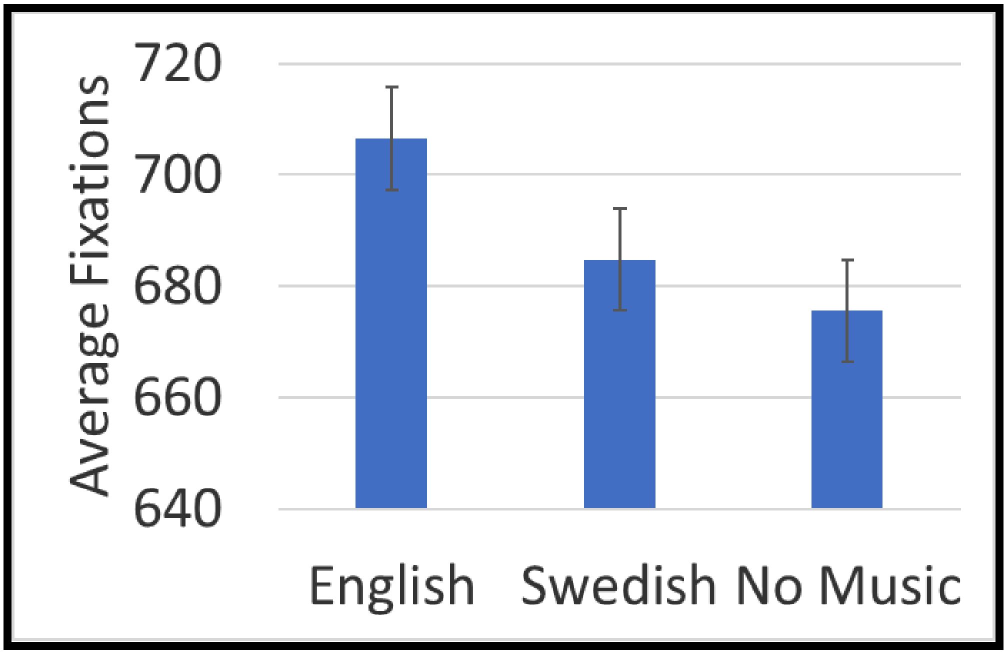

Figure 6. The average number of fixations for all three conditions. Error bars reflect standard error of the mean.

By conducting a one way ANOVA test, we found no significant difference in the average dwell times of participants across the English, Swedish, and no music modalities, $F(2, 92) = 1.709$, $p = 0.187$. This suggests that listening to music, whether in English or Swedish, while reading has no significant effect on the average dwell time of participants.

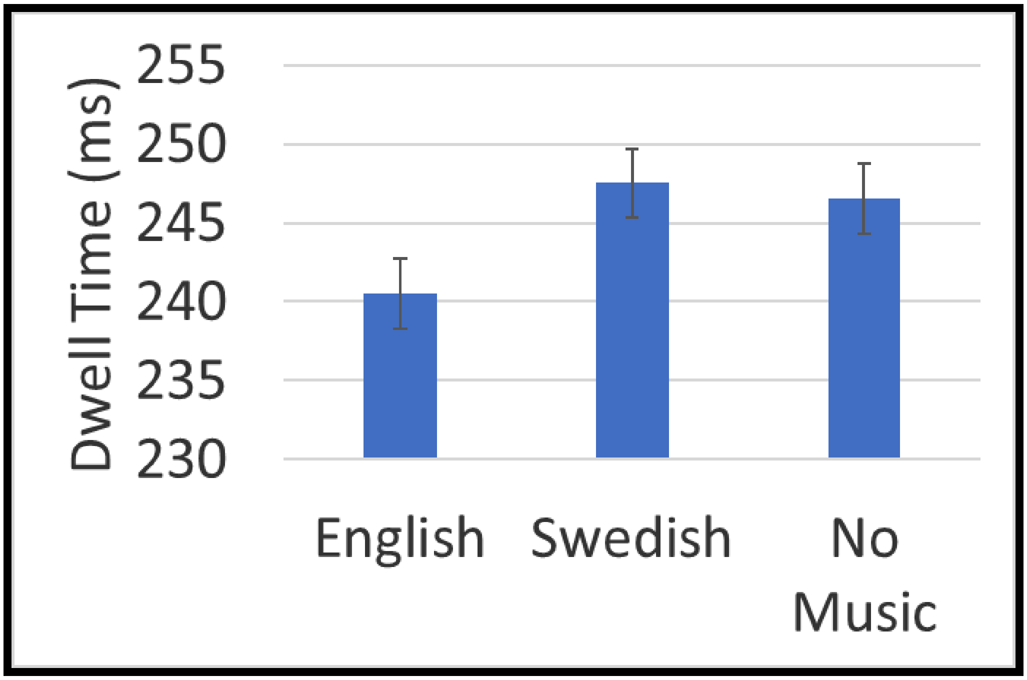

Figure 7. The average dwell time in milliseconds for all three conditions. Error bars reflect standard error of the mean.

By conducting a one way ANOVA test, we found no significant difference in the average blink rates of participants across the English, Swedish, and no music modalities, $F(2, 92) = 0.836$, $p = 0.437$. This suggests that listening to music, whether in English or Swedish, while reading has no significant effect on the average blink rate of participants.

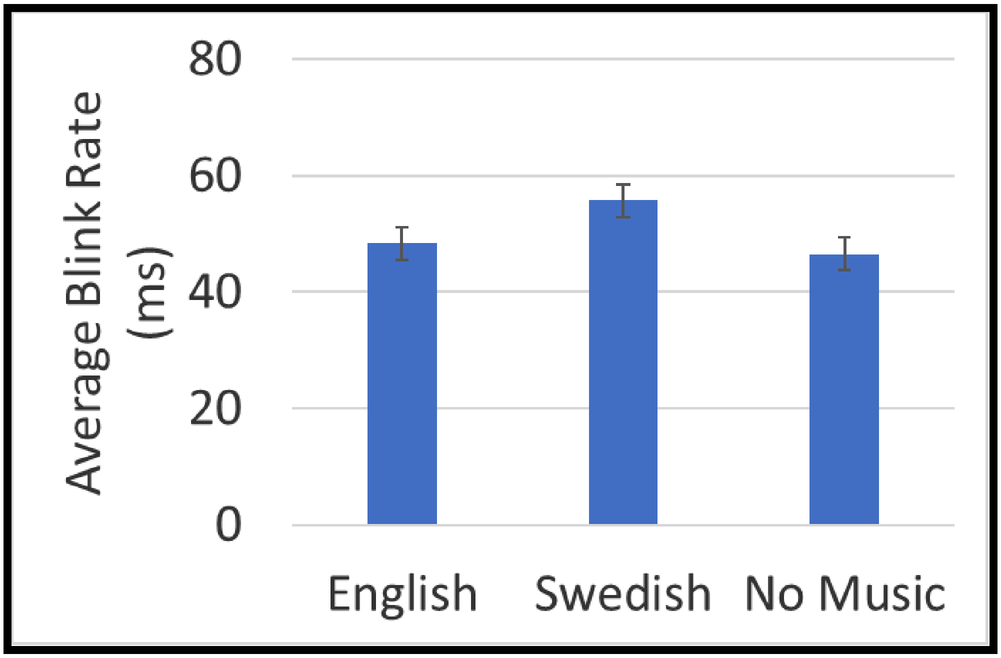

Figure 8. The average blink rate in milliseconds for all three conditions. Error bars reflect standard error of the mean.

Discussion

The present study investigated whether understanding the language of background music would influence reading comprehension during a reading task. Given the commonality with which college students study along with music, the goal of this study was to determine whether the linguistic content of music had a significant effect on cognitive processing–interfering with reading comprehension. While there was no significant difference in reading comprehension accuracy, eye-tracking measures revealed slight differences in visual processing that provide some insight into how background music might influence attention during reading.

When it comes to the eye-tracking data, no significant differences were found for average response time, fixation count, dwell time, or blink rate across all the conditions. However, a difference approaching significance was found for average eye amplitude. In the paired-sample t-tests, a significant difference between English and Swedish music conditions were found, showing that the English music produced greater eye amplitude than the Swedish music. This suggests that comprehensible lyrics may subtly affect attention and engagement while reading.

Previously conducted research as explained in Du et al. (2020) provides support for the idea that background music can influence reading comprehension by providing a different perspective linking behavior to the brain. This study demonstrated that reading with background music increases the neural effort required for semantic integration–reflected by larger N400 amplitudes measured using EEG compared to reading in silence. The N400 component was explained as being associated with processing meaning. Increased amplitude indicated greater difficulty integrating semantic information. In culmination with present findings, this research suggests that background music, in particular music in a comprehensible language, may increase the cognitive and neural cost of reading without necessarily impairing comprehension accuracy.

Limitations

Several limitations were present during the study that should be considered when interpreting the results. First, the quizzes were likely too easy for participants, resulting in near perfect scores for most participants involved. This made it so that we were unable to utilize the quizzes as a marker to determine how much comprehension was affected by music conditions. Second, participants were not given time limits for reading or taking the quizzes, introducing too much variability in time as it was determined by personal preference, and may have minimized detected effects by allowing participants to compensate for potential distractions by taking more time to reread. Additionally, the reading passages differed in emotional content, which may have influenced memory and engagement independently of music condition. For example, as talked about in class, emotionally-charged content has been shown to enhance recall, which may have masked potential effects of distraction. Lastly, the study was limited to two songs from a single genre, which restricts generalizability of the findings to other types of music such as more upbeat or energetic genres like rock, or music without lyrics such as classical music.

Future Directions and Implications

Future research could address these limitations by using more challenging comprehension quizzes similar to the level of academic work expected of the college-age participants, implementing time limits, and selecting emotionally neutral passages. Expanding eye-tracking analysis to include regressions, rereading rates, and word-level measures such as word frequency and length, inspired by the measures used in Zhang et al. (2018), may provide further insight into how background music affects reading processes. For example, rereading and regression rate would allow us to see how often an individual went back in the passage–indicating increased compensation due to difficulty comprehending, and looking at word frequency and length would allow us to compare how well participants comprehended lower frequency, longer words which take more effort to process. Additionally, comparing music with lyrics to instrumental music, or looking at the effects of listening to known versus unknown music may help clarify which characteristics of music are most disruptive to reading. Lastly, considering testing out potential differences in how strong (if present) an effect of distraction (some music condition) has on participants with differing study habits could be interesting: those who study with music and those who study in silence. This would provide further insight into how habits may play into potential variability in attention and reading comprehension.

Conclusion

Ultimately, while background music did not significantly affect reading comprehension accuracy, eye-tracking revealed that the language of background music influenced visual attention during reading. These findings suggest that comprehensible lyrics may subtly interfere with attentional processes even when performance seems unaffected. Despite the previously mentioned limitations, this study has important real-world implications. While many students are under the impression that studying with music has no effect on their performance–or even improves it, current findings suggest that even when comprehension accuracy remains high, listening to music in a familiar language may alter processes related to attention during reading tasks, potentially complicating cognitive efforts. Understanding the more specific ways in which music poses as a potential distractor could help students make more informed decisions when it comes to their studying habits.

References

Abril, C. R., & Flowers, P. J. (2007). *Attention, preference, and identity in music listening by middle school students of different linguistic backgrounds.* Journal of Research in Music Education, 55(3), 204–219. Doi: 10.1177/002242940705500303.

Du, M., Jiang, J., Li, Z., Man, D., & Jiang, C. (2020). *The effects of background music on neural responses during reading comprehension*. Springer Science and Business Media LLC. 10.1038/s41598-020-75623-3.

Furnham, A., & Stephenson, R. (2007). *Musical distractors, personality type and cognitive performance in school children.* Psychology of Music, 35(3), 403–420. https://doi.org/10.1177/0305735607072653.

Purves D, Augustine GJ, Fitzpatrick D, et al., editors. Neuroscience. 2nd edition. Sunderland (MA): Sinauer Associates; 2001. *Types of Eye Movements and Their Functions*. Available from: https://www.ncbi.nlm.nih.gov/books/NBK10991/.

Yıldız, M., Özdemir Cihan, M., & Kurşun, E. (2024). *What is the difference between good and poor young adult readers?* Reading Psychology, 45(4), 362–385. https://doi.org/10.1080/02702711.2024.2308929.

Zhang, H., Miller, K., Cleveland, R., & Cortina, K. (2018). *How listening to music affects reading: Evidence from eye tracking.* Journal of Experimental Psychology: Learning, Memory, and Cognition, 44(11), 1778–1791. 10.1037/xlm0000544 https://research.ebsco.com/c/4oy6qs/viewer/pdf/xpddb3rlan.

Part of Paper Done Together:

Apollo: Procedure

Avery: Results

Katherine: Design and Participants

Rachael: Results

Rebecca: Stimuli